



## LESSON 6-4

## Logarithmic Functions

## Lesson Overview

## Objective

Students will be able to:

- ✓ Graph logarithmic functions and interpret their key features.
- ✓ Write and interpret the inverses of exponential and logarithmic functions.

## Essential Understanding

The inverse relationship between exponential and logarithmic functions reveals key features of the graphs of both functions. Logarithmic functions can be used to model several real-world situations.

## Previously in this topic, students:

- Wrote the inverse of an exponential function using logarithmic notation.

## In this lesson, students:

- Extend their understanding of the inverse relationship between exponential and logarithmic functions by finding key features of logarithmic graphs, as well as interpreting logarithmic models and their inverses.

## Increasing Coherence

Examples 3 and 4 are not necessarily expected to be addressed on high stakes assessments.

## Later in this topic, students will:

- Use their knowledge of properties of exponents, combined with their understanding of the inverse relationship between exponents and logarithms, to identify properties of logarithms.

This lesson emphasizes a blend of *procedural skill and fluency* and *application*.

- Students use key features of logarithmic functions to graph transformations of the parent function.
- Students interpret and compare logarithmic functions as a means for understanding models that relate advertising costs to revenue and models of plane altitudes.



Glossary is available at SavvasRealize.com.

## Vocabulary Builder

## REVIEW VOCABULARY

English | Spanish

- **asymptote** | *asíntota*
- **end behavior** | *comportamiento extremo*
- **reflection** | *reflexión*

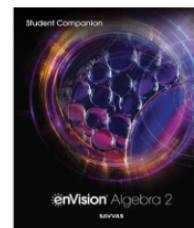
## VOCABULARY ACTIVITY

Review vocabulary by displaying a graph of an exponential function. Have students identify the *domain*, *range*, *x-intercept*, *asymptote* and *end behavior*. If needed, have students complete these sentences.

1. The point at which a graph intersects an axis is the \_\_\_\_\_.  
[intercept]
2. The line that a graph approaches as  $x$  or  $y$  increases in absolute value is a(n) \_\_\_\_\_.  
[asymptote]
3. \_\_\_\_\_ describes the way the graph behaves as  $x$  approaches positive or negative infinity.  
[end behavior]

## Student Companion

Students can do their work for the lesson in their *Student Companion* or on Savvas Realize.



## Mathematics Overview

Students graph logarithmic functions and identify domain, range,  $x$ -intercept, asymptote, and end behavior of the graphs. They graph transformations of logarithmic functions and compare the key features to the graph of the parent function.

Students write and interpret the inverse of a formula involving logarithms. They compare the key features of two logarithmic functions in the context of a real-world situation.

## Applying Math Practices

## Model with Mathematics

Students use equations and graphs to describe and interpret the results of mathematical situations such as advertising costs and airplane altitudes.

## Look For and Make Use of Structure

Students apply general mathematical rules of transforming functions to transform graphs of logarithmic functions. They also use structure by identifying that because logarithmic functions are inverses of exponential functions, their graphs are reflections of exponential functions across the line  $y = x$ .



Activity

## EXPLORE & REASON

**INSTRUCTIONAL FOCUS** Students use their knowledge of graphing and inverses to determine the relationship between the domain and the range of inverse relations. This will enhance students' understanding of the relationship between exponential and logarithmic functions.

**STUDENT COMPANION** Students can complete the *Explore & Reason* activity in their *Student Companion*.

**Before** **WHOLE CLASS**

**Implement Tasks that Promote Reasoning and Problem Solving** **ETP**

**Q:** How are the graphs similar? How are they different?

[All three graphs are parabolas with a vertex at (0, 0), but they open in different directions.]

**During** **SMALL GROUP**

**Support Productive Struggle in Learning Mathematics** **ETP**

**Q:** How do the graphs of the inverse functions  $y = 2x + 4$  and  $y = \frac{1}{2}x - 2$  help you understand this problem?

[The graphs of these two inverse linear functions are reflections of each other across the line  $y = x$ , not across the  $x$ -axis. The same idea can be applied to determining which parabolas are inverses of each other.]

**Q:** How could you represent the domain and the range of each graph in order to compare them?

[Write each domain and range as an inequality.]

**For Early Finishers**

**Q:** Study the graph that is not an inverse of the others. Describe the graph in terms of one of the other two. Then, describe the graph of the inverse of the function whose inverse is not shown.

[The third graph is a reflection of the first graph across the  $x$ -axis. The graph of the inverse is a parabola with the vertex at (0, 0) that opens to the left.]

**After** **WHOLE CLASS**

**Facilitate Meaningful Mathematical Discourse** **ETP**

Ask students to share their answers to Part B.

**Q:** What did you notice about the graphs that were inverses?

[The graphs that are inverses of each other are reflections across the line  $y = x$ .]

### HABITS OF MIND

**Communicate Precisely** How are the points on graphs of functions that are inverses of each other related?

[The  $x$ -values of a function are the  $y$ -values of its inverse, and the  $y$ -values of a function are the  $x$ -values of its inverse.]

Use with EXPLORE & REASON



Explore & Reason is available at SavvasRealize.com.

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### EXPLORE AND REASON

**Compare the graphs.**

A. Which two graphs represent the inverse of each other? Explain

B. **Look for Relationships** What is the relationship between the domain and the range of the two inverse relations?

STUDENT EDITION

**6-4**  
Logarithmic Functions

**EXPLORE & REASON**

**Compare the graphs.**

A. Which two graphs represent the inverse of each other? Explain.

B. **Look for Relationships** What is the relationship between the domain and the range of the two inverse relations?

How is the relationship between logarithmic

SAMPLE STUDENT WORK

- A. The 2 graphs on the left are inverses of each other. Looking at the two graphs, the left graph has points  $(-1, 1)$ ,  $(0, 0)$ , and  $(1, 1)$ , and the middle graph has points  $(1, -1)$ ,  $(0, 0)$ ,  $(1, 1)$ . Since the  $x$ - and  $y$ -values for the two graphs are interchanged, the graphs are inverses of each other.
- B. The domain of the left graph, all real numbers, is the same as the range of the middle graph. The range of the left graph,  $y > 0$ , is equivalent to the domain of the middle graph,  $x > 0$ .

# STEP 2 Understand & Apply



## INTRODUCE THE ESSENTIAL QUESTION

### Establish Mathematics Goals to Focus Learning ETP

Introduce the lesson by helping students recall the inverse relationship between logarithmic and exponential functions. Explain that understanding this relationship will help them to identify and use the key features of the graphs of logarithmic functions.

### EXAMPLE 1 Identify Key Features of Logarithmic Functions

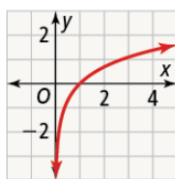
#### Use and Connect Mathematical Representations ETP

**Q:** When graphing  $y = \log_2 x$ , why is it helpful to start by graphing  $y = 2^x$ ?

[The inverse relationship between exponential and logarithmic functions can be used to graph a logarithmic function.]

#### Try It! Answers

1. a.



domain:  $\{x \mid x > 0\}$ ; range: all real numbers; x-intercept: 1; asymptote: y-axis; end behavior: As  $x \rightarrow 0$ ,  $y \rightarrow -\infty$ . As  $x \rightarrow \infty$ ,  $y \rightarrow \infty$ .



Example 1 is Powered by Desmos.

Examples are available at SavvasRealize.com.

**ESSENTIAL QUESTION** How is the relationship between logarithmic and exponential functions revealed in the key features of their graphs?

**EXAMPLE 1** Identify Key Features of Logarithmic Functions

Graph  $g(x) = \log_2 x$ . What are the domain, range, x-intercept, and asymptote? What is the end behavior of the graph?

Logarithmic functions are inverse of exponential functions. Create tables of values for  $f(x) = 2^x$  and  $g(x) = \log_2 x$ .

|        |               |               |   |   |   |
|--------|---------------|---------------|---|---|---|
| $f(x)$ | -2            | -1            | 0 | 1 | 2 |
| $y$    | $\frac{1}{4}$ | $\frac{1}{2}$ | 1 | 2 | 4 |

|        |               |               |   |   |   |
|--------|---------------|---------------|---|---|---|
| $g(x)$ | $\frac{1}{4}$ | $\frac{1}{2}$ | 1 | 2 | 4 |
| $y$    | -2            | -1            | 0 | 1 | 2 |

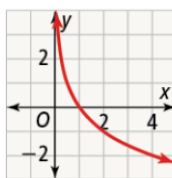
Graph the functions and note the key features of  $g(x)$ .

domain:  $\{x \mid x > 0\}$   
range: all real numbers  
x-intercept: 1  
asymptote: y-axis  
end behavior: As  $x \rightarrow 0$ ,  $y \rightarrow -\infty$ . As  $x \rightarrow \infty$ ,  $y \rightarrow \infty$ .

**Try It!** 1. Graph each function and identify the domain and range. List any intercepts or asymptotes. Describe the end behavior.  
a.  $y = \ln x$   
b.  $y = \log_3 x$

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b.



domain:  $\{x \mid x > 0\}$ ; range: all real numbers; x-intercept: 1; asymptote: y-axis; end behavior: As  $x \rightarrow 0$ ,  $y \rightarrow -\infty$ . As  $x \rightarrow \infty$ ,  $y \rightarrow \infty$ .

## Additional Examples

**Go back**

**ADDITIONAL EXAMPLE 2** **ESSENTIAL QUESTION**

Graph the function  $f(x) = -\log(x + 3) - 4$ .

How does the graph of the function compare to that of the parent function?

Compared to the graph of  $y = \log x$ :

- The value of  $h$  is  $-3$ , so the vertical asymptote is translated 3 units to the left.
- The value of  $a$  is negative, so the graph is reflected over the y-axis.
- The value of  $k$  is  $-4$ , so the graph is translated 4 units down.

**SOLUTION**

**Example 2** Students will graph logarithmic functions with both translations and reflections with this additional example.

**Q:** How do the vertical asymptote and the x-intercept of  $f(x)$  compare to the vertical asymptote and the x-intercept of  $g(x) = \log(x + 3) + 4$ ?

[They are the same for both functions.]

**Additional Examples** are available at SavvasRealize.com.

**Go back**

**ADDITIONAL EXAMPLE 3** **ESSENTIAL QUESTION**

Find the inverse of the function.

$f(x) = 11(4)^{x-6} + 9$

$f(x) = 11(4)^{x-6} + 9$   
 $y = 11(4)^{x-6} + 9$   
 $x = 11(4)^{y-6} + 9$   
 $x - 9 = 11(4)^{y-6}$   
 $\frac{x-9}{11} = (4)^{y-6}$   
 $y - 6 = \log_4\left(\frac{x-9}{11}\right)$   
 $y = \log_4\left(\frac{x-9}{11}\right) + 6$

The inverse of  $f(x)$  is  $f^{-1}(x) = \log_4\left(\frac{x-9}{11}\right) + 6$ .

**SOLUTION**

**Example 3** Students will find the inverse of an exponential function using logarithms with this additional example.

**Q:** How does knowing that logarithm functions and exponential functions are inverses help you solve this problem?

[Knowing that they are inverses helps you see an overall direction for solving if you get stuck after you exchange the  $x$  and  $y$  variables.]



## EXAMPLE 2

### Graph Transformations of Logarithmic Functions

#### Use and Connect Mathematical Representations ETP

**Q:** How is graphing transformations of logarithmic functions the same as graphing transformations of exponential functions? How is it different?

[same:  $h = -3$  implies a shift to the left; different:  $h$  causes the shift in the asymptote of a logarithmic function, whereas  $k$  causes the shift of the asymptote in an exponential function.]

#### Try It! Answers

2. a. translated up 4 units
- b. vertical stretch of 5

#### HABITS OF MIND

**Use Structure** Does the graph of either  $y = \ln x + 4$  or  $y = \ln(x + 4)$  have an intercept that is different from the intercept of  $y = \ln x$ ? Explain.

[The  $x$ -intercept of  $y = \ln x + 4$  is different from the  $x$ -intercept of  $y = \ln x$ , and  $y = \ln x + 4$  does not have a  $y$ -intercept. The  $x$ -intercept of  $y = \ln(x + 4)$  is different from the  $x$ -intercept of  $y = \ln x$  and  $y = \ln(x + 4)$  has a  $y$ -intercept.]

## EXAMPLE 3

### Inverses of Exponential and Logarithmic Functions

#### Support Productive Struggle in Learning Mathematics ETP

**Q:** Why does a horizontal shift in an exponential equation become a vertical shift in its inverse.

[With inverse functions, the  $x$  and  $y$  values are interchanged. So a transformation that affected the  $x$  values of the original function would affect the  $y$  values of its inverse.]

#### Try It! Answers

3. a.  $f^{-1}(x) = \log_3 x - 2$
- b.  $g^{-1}(x) = 7^{x+2}$

### Support Struggling Students

**USE WITH EXAMPLE 2** Some students may have difficulty comparing the graph of the function to the parent function.

- Have students practice finding the  $x$ -intercept or the asymptote of each function.

**Q:** Find the  $x$ -intercept of  $g(x) = \log(x - 5)$ . [6]

**Q:** Find the asymptote of  $h(x) = \log(x + 3)$ .  
[ $x = -3$ ]

**Q:** How does the graph of  $h(x) = \log(x + 3)$  compare to the parent function  $f(x) = \log x$ ?  
[The graph of  $h$  is shifted 3 units to the left of the graph of  $f$ .]



Examples are available at SavvasRealize.com.

## EXAMPLE 2

### Graph Transformations of Logarithmic Functions

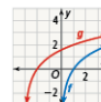
Graph the function  $g(x) = \log_2(x + 3)$ . How do the asymptote and  $x$ -intercept of the given function compare to those of the parent function  $f(x) = \log_2 x$ ?

$$f(x) = \log_2 x$$

$$g(x) = \log_2(x + 3) = f(x - (-3))$$

Since  $g(x) = f(x + 3)$ , then  $g$  is the logarithmic function translated 3 units to the left.

The vertical asymptote and the  $x$ -intercept each shift 3 units to the left.



**Try It!** 2. Describe how each graph compares to the graph of  $f(x) = \ln x$ .

- a.  $g(x) = \ln x + 4$
- b.  $h(x) = 5 \ln x$

#### CONCEPTUAL UNDERSTANDING

**LOOK FOR RELATIONSHIPS**  
 $f(x)$  is a translation of the parent function  $y = 10^x$  one unit left.  $f^{-1}(x)$  is a translation of the parent function  $y = \log x$  one unit down. Graphical translations of a function and its inverse are directly related. Explain.

## EXAMPLE 3

### Inverses of Exponential and Logarithmic Functions

What is the equation of the inverse of the functions?

A.  $f(x) = 10^{x+1}$

$$y = 10^{x+1}$$

$$x = 10^{y+1}$$

$$y + 1 = \log x$$

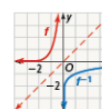
$$y = \log x - 1$$

The equation of the inverse of  $f(x) = 10^{x+1}$  is  $f^{-1}(x) = \log x - 1$ .

Verify by function composition.

$$\begin{aligned} f(f^{-1}(x)) &= 10^{(\log x - 1) + 1} \\ &= 10^{\log x} \\ &= x \end{aligned}$$

$$\begin{aligned} f^{-1}(f(x)) &= \log 10^{x+1} - 1 \\ &= x + 1 - 1 \\ &= x \end{aligned}$$



B.  $g(x) = \log_7(x + 5)$

$$y = \log_7(x + 5)$$

$$x = \log_7(y + 5)$$

$$y + 5 = 7^x$$

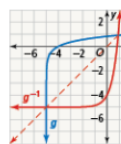
$$y = 7^x - 5$$

The equation of the inverse of  $g(x) = \log_7(x + 5)$  is  $g^{-1}(x) = 7^x - 5$ .

Verify by function composition.

$$\begin{aligned} g(g^{-1}(x)) &= \log_7(7^x - 5 + 5) \\ &= \log_7 7^x \\ &= x \end{aligned}$$

$$\begin{aligned} g^{-1}(g(x)) &= 7^{\log_7(x+5)} - 5 \\ &= x + 5 - 5 \\ &= x \end{aligned}$$



### Extend Student Thinking

**USE WITH EXAMPLE 3** Have students explore finding the inverse of functions with two translations.

- Have students find the inverse of each function.

1.  $f(x) = 2^{x+3} - 3$  [ $f^{-1}(x) = \log_2(x + 3) - 3$ ]

2.  $g(x) = \log_4(x - 4) - 1$  [ $g^{-1}(x) = 4^{x+1} + 4$ ]

3.  $h(x) = -\log(x + 1) + 6$  [ $h^{-1}(x) = 10^{6-x} - 1$ ]

**Q:** How are the graphs of these functions and their inverses related?

[The graphs of the inverses are reflections of the original function across the line  $y = x$ .]



# STEP 2 Understand & Apply



## EXAMPLE 4 Use a Logarithmic Function as a Model

### Pose Purposeful Questions ETP

**Q:** Why does it make sense that the function is not defined for negative values of  $t$ ?

[Negative values of  $t$  correspond to times before takeoff, which is when the plane is constantly on the ground.]

### Try It! Answers

4. about 21 minutes; The model is not accurate past 21 minutes since the logarithmic function continues to increase above the cruising altitude.

### HABITS OF MIND

Use with EXAMPLES 3 & 4

**Reason** Does it make sense that the plane is at 31,528 feet after 10 minutes but only 35,000 feet after 15.14 minutes?

[Yes, because the rate of increase of a logarithmic function decreases as  $t$  increases.]

Examples are available at SavvasRealize.com.

EXAMPLE 3 CONTINUED

**Try It!** 3. Find the inverse of each function. Then verify they are inverses by composition.

a.  $f(x) = 3^{x+2}$

b.  $g(x) = \log_7 x - 2$

### EXAMPLE 4 Use a Logarithmic Function as a Model

A logarithmic function can approximate the altitude of a plane over time.

**A.** What is the altitude of the plane 10 minutes after takeoff?

Substitute 10 for  $t$  in the equation of the function.

$$\begin{aligned} A &= 9,200 \ln 10 + 10,000 \\ &\approx 9,200(2.303) + 10,000 \\ &\approx 21,188 + 10,000 \\ &\approx 31,188 \end{aligned}$$

After 10 minutes, the plane's altitude is about 31,188 feet.

**B.** How long after takeoff will the plane reach an altitude of 35,000 feet? Rearrange the equation to isolate the variable  $t$ .

$$A = 9,200 \ln t + 10,000$$

$$A - 10,000 = 9,200 \ln t$$

$$\frac{A - 10,000}{9,200} = \ln t$$

$$t = e^{\left(\frac{A - 10,000}{9,200}\right)}$$

In the rearranged equation, time is a function of altitude, so use substitution to find time.

$$\begin{aligned} t &= e^{\left(\frac{35,000 - 10,000}{9,200}\right)} \\ &\approx 15.14 \end{aligned}$$

The plane will reach an altitude of 35,000 feet after about 15.14 minutes.

**C.** Evaluate the model against the altitude of the plane.

On the domain interval  $(0, 0.337)$  the logarithmic function is negative. The altitude of the plane is not correct during this interval. Also, the plane starts at the origin, which is on the asymptote of the model.

**Try It!** 4. Commercial airplanes usually fly no higher than 38,000 ft. How long will it take the plane to reach this altitude? How does a maximum cruising height compare with the model?

### STUDY TIP

The original equation is a logarithmic function, while the rearranged equation is an exponential function. The exponential function is the inverse of the logarithmic function.

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## ELL English Language Learners (Use with EXAMPLE 3)

**SPEAKING BEGINNING** In small groups, have students discuss the meanings of *inter-* and *change*, and how they relate to the meaning of *interchange*.

**Q:** Where do you hear the word *interchange* used?

[Sample: when merging from one highway to another]

**Q:** What does the word *interchange* mean? [a mutual, or reciprocal, exchange]

**LISTENING DEVELOPING** Have students listen to the definition of *inverse* and answer the question.

The definition of *inverse* is the opposite or reverse of something.

**Q:** What does *inverse* mean in math?

[opposite operations such as addition and subtraction; exchanging  $x$  and  $y$  in functions]

**WRITING EXPANDING** Have one student hold a card with X on it and another student hold a card with Y on it. Ask them to come to the front of the room and exchange the cards.

**Q:** What word from the example was just demonstrated? [interchange]

**Q:** Write definitions for *interchange* and *inverse* in your own words. Explain how the two words are related. [Check students' work.]



## CONCEPT SUMMARY Logarithmic Functions

**Q:** How are the domain and the range of  $y = \log x$  related to the domain and the range of  $y = 10^x$ ?

[The domain of  $y = \log x$  is the same as the range of  $y = 10^x$  and the range of  $y = \log x$  is the same as the domain of  $y = 10^x$ .]

**Q:** How are the  $x$ -intercept and the asymptote of  $y = \log x$  affected if the graph is translated 3 units to the right?

[The  $x$ -intercept and asymptote are also translated 3 units to the right. The  $x$ -intercept is 4 and the asymptote is  $x = 3$ .]

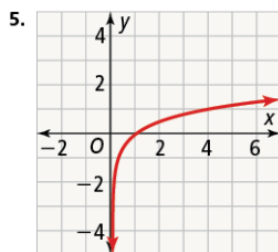
## Do You UNDERSTAND? | Do You KNOW HOW?

### Common Error

**Exercise 5** Students may ignore the  $-1$  before adding  $1\frac{1}{2}$ , writing the equation as  $g(x) = \ln x + 1\frac{1}{2}$ . Instead of  $g(x) = \ln x + \frac{1}{2}$ . Explain that the graph of  $f(x) = \ln x - 1$  is the graph of  $h(x) = \ln x$  shifted down 1 unit, so shifting this function up by  $1\frac{1}{2}$  units is the same as shifting  $h(x)$  up  $\frac{1}{2}$  unit.

## Answers

- The domain of the logarithmic function is the same as the range of the exponential function, and the range of the logarithmic function is the same as the domain of the exponential function. The logarithmic function has an  $x$ -intercept of 1, while the exponential function has a  $y$ -intercept of 1.
- This is the range of the function. The domain is  $\{x \mid x > 0\}$ .
- The graph of  $g(x)$  is the reflection of the graph of  $f(x)$  across the  $x$ -axis.
- Rewrite the problem using  $x$  and  $y$ , then exchange the places of the  $x$  and  $y$ . Next solve for the  $y$ . In this step use the relationship between exponential and logarithmic functions. Finally, rewrite in inverse function notation.  $h^{-1}(x) = 6^x - 4$ .



domain:  $\{x \mid x > 0\}$ ; range: all real numbers;  $x$ -intercept: 1; asymptote:  $y$ -axis; end behavior: As  $x \rightarrow 0$ ,  $y \rightarrow -\infty$ .  
As  $x \rightarrow \infty$ ,  $y \rightarrow \infty$ .

- $g(x) = \ln x + \frac{1}{2}$
- $x = e^{\frac{y}{5}} - 1$ ; This function gives  $x$ , the number of minutes after the release of a song, in terms of the number of downloads in hundreds,  $y$ .



Concept Summary, Do You Understand, and Do You Know How are available at SavvasRealize.com.

## CONCEPT SUMMARY Logarithmic Functions

|                                                                                                                                                   |                                                                                                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| <b>GRAPH</b><br><p>The functions are inverses so their graphs are reflections of each other across the line with equation <math>y = x</math>.</p> |                                                                                                      |
| <b>EQUATIONS</b><br>$y = \log x$                                                                                                                  | $y = 10^x$                                                                                           |
| <b>KEY FEATURES</b><br>Domain: $\{x \mid x > 0\}$<br>Range: all real numbers<br>$x$ -intercept: 1<br>Asymptote: $y$ -axis                         | Domain: all real numbers<br>Range: $\{y \mid y > 0\}$<br>$y$ -intercept: 1<br>Asymptote: $x$ -axis   |
| <b>END BEHAVIOR</b><br>As $x \rightarrow 0$ , $y \rightarrow -\infty$<br>As $x \rightarrow \infty$ , $y \rightarrow \infty$                       | As $x \rightarrow -\infty$ , $y \rightarrow 0$<br>As $x \rightarrow \infty$ , $y \rightarrow \infty$ |

### Do You UNDERSTAND?

- ESSENTIAL QUESTION** How is the relationship between logarithmic and exponential functions revealed in the key features of their graphs?
- Error Analysis** Raynard claims the domain of the function  $y = \log_3 x$  is all real numbers. Explain the error Raynard made.
- Communicate Precisely** How are the graphs of  $f(x) = \log_5 x$  and  $g(x) = -\log_5 x$  related?
- Use Structure** Explain the necessary steps to find the inverse of  $h(x) = \log_6(x + 4)$ .

### Do You KNOW HOW?

- Graph the function  $y = \log_4 x$  and identify the domain and range. List any intercepts or asymptotes. Describe the end behavior.
- Write the equation for the function  $g(x)$ , which can be described as a vertical shift  $\frac{1}{2}$  units up from the function  $f(x) = \ln x - 1$ .
- The function  $y = 5 \ln(x + 1)$  gives  $y$ , the number of downloads, in hundreds,  $x$  minutes after the release of a song. Find the equation of the inverse and interpret its meaning.



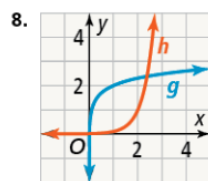
- Sketch the functions represented by the tables. Identify which graph is the logarithmic function. Are the two functions inverses?

|        |            |      |   |       |    |
|--------|------------|------|---|-------|----|
| $x$    | $10^{-10}$ | 0.01 | 1 | 2     | 10 |
| $g(x)$ | -8         | 0    | 2 | 2.301 | 3  |

|        |       |      |     |   |    |     |
|--------|-------|------|-----|---|----|-----|
| $x$    | -1    | 0    | 1   | 2 | 3  | 4   |
| $h(x)$ | 0.001 | 0.01 | 0.1 | 1 | 10 | 100 |

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Yes they are inverses.

- no; The  $x$ -intercept of the blue function is not equal to the  $y$ -intercept of the red function.
- $g(x)$  is shifted 7 units to the right; for each  $x$ ,  $g(x) = f(x - 7)$ , so the  $y$ -values of the functions are equal for  $x$ -values that are 7 units apart.
- When rewriting in logarithmic form, the student should have placed the expression  $x - 2$  in parenthesis. The correct inverse function is  $f^{-1}(x) = \log_5(x - 2) + 6$ .
- $w = \log_{1.6} m - 2$ ; the number of weeks after opening when the number of members reaches  $m$
- $g(x) = \log_3(x - 2) + 1$



## PRACTICE & PROBLEM SOLVING



Practice & Problem Solving and video tutorials are available at SavvasRealize.com.

Scan the QR code to access a video tutorial.

SavvasRealize.com



**Lesson Practice** You may opt to have students complete the automatically scored Practice and Problem Solving items online powered by MathXL for School.

Choose from: **Lesson Practice**

**Adaptive Practice**

You may also take advantage of the bank of exercises for assigning additional practice.

### Assignment Guide

| On-Level    | Advanced    |
|-------------|-------------|
| 7–22, 25–31 | 7–12, 15–31 |

### Engage Through Student Choice

Promote student agency by allowing students to choose practice items. You may structure this choice in many ways.

For example:

Assign each section a point value. Students choose at least one item from each section and items chosen should have a minimum of 20 total points.

**Understand, Apply**.....2 points each  
**Practice**.....1 point each  
**Assessment Practice**.....1 point each  
**Performance Task** .....3 points

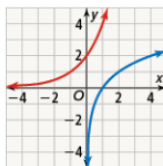
### Item Analysis

| Example | Items            | DOK |
|---------|------------------|-----|
| 1       | 12–15            | 1   |
| 2       | 8, 11, 16–18, 29 | 2   |
| 3       | 19–24, 30        | 1   |
|         | 7, 9             | 2   |
| 4       | 10, 25, 27, 31   | 2   |
| 5       | 26               | 2   |
|         | 28               | 3   |

## PRACTICE & PROBLEM SOLVING

### UNDERSTAND

9. **Look for Relationships** Are the logarithmic and exponential functions shown inverses of each other? Explain.



10. **Communicate Precisely** How is the graph of the logarithmic function  $g(x) = \log_2(x - 7)$  related to the graph of the function  $f(x) = \log_2 x$ ? Explain your reasoning.

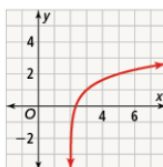
11. **Error Analysis** Describe and correct the error a student made in finding the inverse of the exponential function  $f(x) = 5^{x-6} + 2$ .

$y = 5^{x-6} + 2$   
 $x = 5^{y-6} + 2$   
 $x - 2 = 5^{y-6}$   
 $y - 6 = \log_5 x - 2$   
 $y = \log_5 x - 2 + 6$   
 $y = \log_5 x + 4$   
 $f^{-1}(x) = \log_5 x + 4$

Write in  $y = f(x)$  form.  
 Interchange  $x$  and  $y$ .  
 Subtract 2 from each side.  
 Rewrite in logarithmic form.  
 Add 6 to each side.  
 Simplify.

12. **Make Sense and Persevere** The number of members  $m$  who joined a new workout center  $w$  weeks after opening is modeled by the equation  $m = 1.6^w + 2$ , where  $0 \leq w \leq 10$ . Find the inverse of the function and explain what the inverse tells you.

13. **Use Structure** The graph shows a transformation of the parent graph  $f(x) = \log_3 x$ . Write an equation for the graph.



### PRACTICE

Graph each function and identify the domain and range. List any intercepts or asymptotes. Describe the end behavior. SEE EXAMPLE 1

14.  $y = \log_5 x$       15.  $y = \log_8 x$   
 16.  $y = \log_{\frac{3}{10}} x$       17.  $y = \log_{0.1} x$

Describe the graph in terms of transformations of the parent function  $f(x) = \log_6 x$ . Compare the asymptote and  $x$ -intercept of the given function to the parent function. SEE EXAMPLE 2

18.  $g(x) = \frac{1}{2} \log_6 x$       19.  $g(x) = \log_6(-x)$

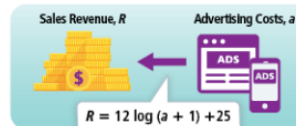
20. Describe how the graph of  $g(x) = -\ln(x + 0.5)$  is related to the graph of  $f(x) = \ln x$ . SEE EXAMPLE 2

Find the equation of the inverse of each function. SEE EXAMPLE 3

21.  $f(x) = 5^{x-3}$       22.  $f(x) = \left(\frac{1}{2}\right)^{x-1}$   
 $f^{-1}(x) = \log_5 x + 3$        $f^{-1}(x) = \log_{\frac{1}{2}} x + 1$   
 23.  $f(x) = 6^{x+7}$       24.  $f(x) = \log_2(8x)$   
 $f^{-1}(x) = \log_6 x - 7$        $f^{-1}(x) = \frac{x}{8}$   
 25.  $f(x) = \ln(x + 3) - 1$       26.  $f(x) = 4 \log_2(x - 3) + 2$   
 $f^{-1}(x) = e^{x+1} - 3$        $f^{-1}(x) = 2^{\frac{x-2}{4}} + 3$

27. The altitude  $y$ , in feet, of a plane  $t$  minutes after takeoff is approximated by the function  $y = 5,000 \ln(0.05t) + 8,000$ . Solve for  $t$  in terms of  $y$ . What is a situation in which it would be easier to use your new equation rather than the original? SEE EXAMPLE 4

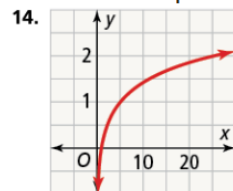
28. A company uses this function to relate sales revenue,  $R$ , and advertising costs,  $a$ . What is the equation of the inverse of the equation? Which equation would be easier to use to find a value of  $a$  for a particular value of  $R$ ? SEE EXAMPLE 4



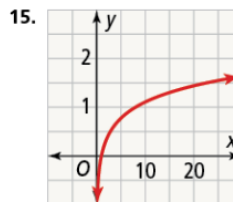
LESSON 6-4 Logarithmic Functions 313

## Answers

9–13. Answers on previous page.



domain:  $\{x \mid x > 0\}$ ; range: all real numbers; intercept:  $x$ -intercept 1; asymptote:  $y$ -axis; end behavior: As  $x \rightarrow 0$ ,  $y \rightarrow -\infty$ . As  $x \rightarrow \infty$ ,  $y \rightarrow \infty$ .

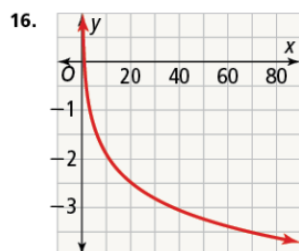


domain:  $\{x \mid x > 0\}$ ; range: all real numbers;  $x$ -intercept 1; asymptote:  $y$ -axis; As  $x \rightarrow 0$ ,  $y \rightarrow -\infty$ . As  $x \rightarrow \infty$ ,  $y \rightarrow \infty$ .

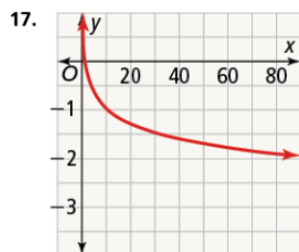
See answers for 16–20, 27, 28 on the next page.



## Answers



domain:  $\{x \mid x > 0\}$ ; range: all real numbers; intercept: x-intercept 1; asymptote: y-axis; end behavior: As  $x \rightarrow 0$ ,  $y \rightarrow \infty$ . As  $x \rightarrow \infty$ ,  $y \rightarrow -\infty$ .



domain:  $\{x \mid x > 0\}$ ; range: all real numbers; intercept: x-intercept 1; asymptote: y-axis; end behavior: As  $x \rightarrow 0$ ,  $y \rightarrow \infty$ . As  $x \rightarrow \infty$ ,  $y \rightarrow -\infty$ .

18. vertical shrink of  $\frac{1}{2}$ ; asymptote: y-axis (same as parent function); x-intercept 1 (same as parent function)

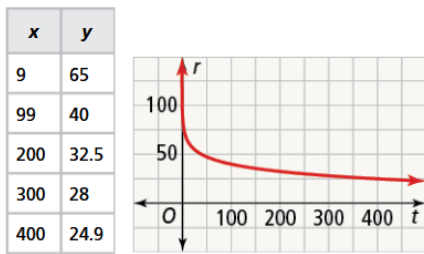
19. reflection over the y-axis; asymptote: y-axis (same as parent function); x-intercept -1 (different from parent function)

20. Reflection of  $f(x)$  shifted 0.5 units to the left.

27.  $t = 20e^{\frac{y-8,000}{5,000}}$ ; Use this new equation when the altitude of the plane is known but the time since takeoff is not.

28. The inverse of the formula is  $a = 10^{\frac{R-25}{12}} - 1$ . It would be easier to use the inverse to find  $a$  when  $R$  is known.

29. a. Check students' work.



b.  $t = 10^{\frac{r-90}{-25}} - 1$ ; the number of months since taking the course at which a student has a retention,  $r$



Practice & Problem Solving is available at SavvasRealize.com.

## PRACTICE & PROBLEM SOLVING

### APPLY

29. **Model with Mathematics** The equation  $r = 90 - 25 \log(t + 1)$  is to model a student's retention  $r$  after taking a physics course where  $r$  represents a student's test score (as a percent), and  $t$  represents the number of months since taking the course.

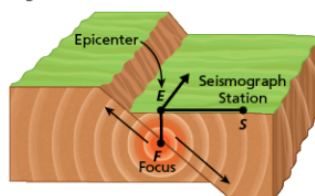
- a. Make a table of values for ordered pairs that represent  $r = 90 - 25 \log(t + 1)$ , rounding to the nearest tenth. Then sketch the graph of the function on a coordinate plane through those ordered pairs. (You may use a graphing calculator to check.)

- a. Find the equation of the inverse. Interpret the meaning of this function.

30. **Higher Order Thinking** As shown by the diagram, an earthquake occurs below Earth's surface at point  $F$  (the focus). Point  $E$ , on the surface above the focus, is called the *epicenter*. A seismograph station at point  $S$  records the waves of energy generated by the earthquake. The surface wave magnitude  $M$  of the earthquake is given by this formula:

$$M = \log\left(\frac{A}{T}\right) + 1.66(\log D) + 3.3$$

In the formula,  $A$  is the amplitude of the ground motion in micrometers,  $T$  is the period in seconds, and  $D$  is the measure of  $ES$  in degrees.



- a. Find surface wave magnitude of an earthquake with  $A = 700$  micrometers,  $T = 2$  and  $D = 100^\circ$ .
- b. In the formula,  $20^\circ < D \leq 160^\circ$ . By how much can the size of arc  $ES$  affect the surface wave magnitude? Explain.

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30. a. magnitude 9.2

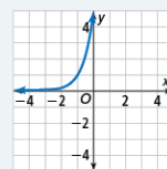
- b. The size of the arc can add between 2.2 and 3.7 to the surface wave magnitude, meaning the arc size can affect the surface wave magnitude by up to 1.5.

### ASSESSMENT PRACTICE

31. The logarithmic function  $g(x) = \ln x$  is transformed to  $h(x) = \ln(x + 2) - 1$ . Which of the following are true? Select all that apply.

- ☐  $g(x)$  is translated 2 units upward.  
☐  $g(x)$  is translated 2 units to the right.  
☒  $g(x)$  is translated 2 units to the left.  
☒  $g(x)$  is translated 1 unit downward.  
☐  $g(x)$  is translated 1 unit to the left.  
☒ The vertical asymptote shifts 2 units to the left.  
☐ The vertical asymptote shifts 2 units to the right.

32. **SAT/ACT** The graph shows the exponential function  $f(x) = 5^{x+1}$ . Which of the following functions represents its inverse,  $f^{-1}(x)$ ?



- ☐ A  $f^{-1}(x) = 1 + \log_5 x$     ☐ B  $f^{-1}(x) = \log_5 x - 1$   
☒ C  $f^{-1}(x) = \log_5(x - 1)$     ☐ D  $f^{-1}(x) = \log_5(x + 1)$

33. **Performance Task** The logarithmic function  $M(d) = 5 \log d + 2$  is used to find the limiting magnitude of a telescope, where  $d$  represents the diameter of the lens of the telescope (mm) that is being used for the observation.

**Part A** Find the limiting magnitude of a telescope having a lens diameter of 40 mm.

**Part B** Find the equation of the inverse of this function.

**Part C** Interpret why astronomers may wish to use the inverse of this function. Justify your reasoning.

**Part D** Using the inverse function, find the diameter of the lens that has a limiting magnitude of 13.5. Check your answer with the table function of your graphing calculator.

33. a. magnitude 10

- b.  $d = 10^{\frac{M-2}{5}}$   
c. Astronomers can use this function when the limiting magnitude of a telescope is known to find the diameter of the lens of the telescope.  
d. 199.5 mm





## LESSON QUIZ

Use the Lesson Quiz to assess students' understanding of the mathematics in the lesson.

Students can take the Lesson Quiz online or you can download a printable copy from [SavvasRealize.com](https://www.savvasrealize.com). The Lesson Quiz is also available in the *Assessment Resources* book.

### Item Analysis

| Item | DOK | Skills Review and Practice |
|------|-----|----------------------------|
| 1    | 2   | F69, F70                   |
| 2    | 2   | F69, F70                   |
| 3    | 2   | F70                        |
| 4    | 2   | F70                        |
| 5    | 2   | F70                        |



Use the student scores on the Lesson Quiz to prescribe differentiated assignments.

If students take the Lesson Quiz online, it will be automatically scored and appropriate differentiated practice will be assigned based on student performance.

|                       |            |                                                                                                                                                                    |
|-----------------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>I</b> Intervention | 0–3 points | <ul style="list-style-type: none"> <li>Reteach to Build Understanding</li> <li>Mathematical Literacy and Vocabulary</li> <li>Lesson Virtual Nerd videos</li> </ul> |
| <b>O</b> On-Level     | 4 points   | <ul style="list-style-type: none"> <li>Enrichment</li> </ul>                                                                                                       |
| <b>A</b> Advanced     | 5 points   | <ul style="list-style-type: none"> <li>Enrichment</li> </ul>                                                                                                       |



Lesson Quiz is available at [SavvasRealize.com](https://www.savvasrealize.com).

## ASSESSMENT RESOURCES

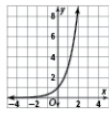
Name \_\_\_\_\_

enVision Algebra 2  
SavvasRealize.com

### 6-4 Lesson Quiz

Logarithmic Functions

- The graph shows the function  $f(x) = 3^x$ .



What is the value of  $f^{-1}(x)$  at  $x = 3$ ?

$f^{-1}(3) =$
- The logarithmic function  $f(x) = \log x$  is transformed to  $g(x) = \log(x + 1) + 3$ . Which statement is true?

☐ A The graph of  $f(x)$  is translated 1 unit upward.

☐ B The graph of  $f(x)$  is translated 3 units downward.

☐ C The vertical asymptote shifts from  $x = 0$  to  $x = -1$ .

☐ D The vertical asymptote shifts from  $x = -1$  to  $x = 3$ .
- Complete the equation of the inverse of the function  $f(x) = \log_2(9x)$ .

☒ A  $f^{-1}(x) = \frac{1}{9}(2^x)$

☐ B  $f^{-1}(x) = \frac{1}{2}(9^x)$

☐ C  $f^{-1}(x) = 9(2^x)$

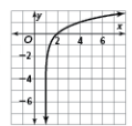
☐ D  $f^{-1}(x) = \log_9(2x)$
- An internet service provider uses the function  $R = 3 \log(a + 2) + 15$  to relate  $R$ , its sales revenue from hosting websites, to  $a$ , the cost of advertising. Solve for  $a$  in terms of  $R$ .

☐ A  $a = \frac{R - 15}{3} - 2$

☐ B  $a = \frac{1}{10} \left( \frac{R - 15}{3} - 2 \right)$

☒ C  $a = 10^{\frac{R - 15}{3}} - 2$

☐ D  $a = 10^{\frac{R - 15}{3}} + 2$
- Consider  $f(x) = 3x - 1$  and the graph of  $g(x)$  shown.



Select the function that makes the statement true.

|                                             | $f(x)$                              | $g(x)$                              | Both                                |
|---------------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|
| Has the greater $x$ -intercept              | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
| Has an asymptote of $y = -1$                | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            |
| Is always increasing                        | <input type="checkbox"/>            | <input type="checkbox"/>            | <input checked="" type="checkbox"/> |
| Is positive over the interval $(2, \infty)$ | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |

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## DIFFERENTIATED RESOURCES

**I** = Intervention **O** = On-Level **A** = Advanced

### Reteach to Build Understanding **I**

Provides scaffolded reteaching for the key lesson concepts.

#### MathXL® for School

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### 6-4 Reteach to Build Understanding

Logarithmic Functions

- All exponential functions of the form  $f(x) = b^x$  and logarithmic functions of the form  $f(x) = \log_b(x)$  are **inverse**, because these functions have exactly one  $x$ -value for each  $y$ -value and **exactly one**  $y$ -value for each  $x$ -value.
- Find and correct the error a student made when finding the inverse of the logarithmic function  $f(x) = \log_4(x+2) - 5$ .
 

$y = \log_4(x+2) - 5$  Write in  $y = f(x)$  form.  
 $x = \log_4(y+2) - 5$  Interchange  $x$  and  $y$ .  
 $x + 5 = \log_4(y+2)$  Add 5 on each side.  
 $4^{x+5} = y+2$  Rewrite in exponential form.  
 $4^{x+5} - 2 = y$  Subtract 2 from each side.  
 $4^{x+5} - 2 = y$  Divide by 4 on each side.

The equation of the inverse of  $f(x) = \log_4(x+2) - 5$  is  $f^{-1}(x) = \frac{4^{x+5}-2}{4}$ .

3. The  $f(x) = 3^{x-1} - 1$  and  $g(x) = \log_3(x+1) + 2$  are inverse functions shown on the graph at the right. Complete the table without using a calculator.

| $x$ | $f(x)$ | $g(f(x))$ |
|-----|--------|-----------|
| 0   | -1     | 0         |
| 1   | 0      | 1         |
| 2   | 1      | 2         |
| 3   | 2      | 3         |
| 4   | 3      | 4         |
| 5   | 4      | 5         |

Domain: All real numbers Domain:  $x \geq -1$   
 Range:  $y > -1$  Range: All real numbers  
 Asymptote:  $y = -1$  Asymptote:  $x = -1$   
 Int. Point:  $(0, -1)$  Int. Point:  $(-1, 0)$   
 All:  $f(g(x)) = x$  All:  $g(f(x)) = x$

### Additional Practice **I O**

Provides extra practice for each lesson.

#### MathXL® for School

Name \_\_\_\_\_

### 6-4 Additional Practice

Logarithmic Functions

Graph the function below and identify the domain, range,  $x$ -intercept,  $y$ -intercept, asymptote, and end behavior. Compare the graph to the parent function.

1.  $f(x) = \log_4(x-2) + 2$   
 domain:  $\{x | x > 2\}$   
 range: all real numbers  
 $x$ -intercept:  $2\frac{1}{6}$   
 $y$ -intercept: none  
 asymptote:  $x = 2$   
 end behavior: As  $x \rightarrow 2$ ,  $y \rightarrow -\infty$   
 As  $x \rightarrow \infty$ ,  $y \rightarrow \infty$   
 The graph is shifted right 2 units and up 2 units.  
 Find the inverse of each function.

2.  $f(x) = 6 \log_2(x-6)$  3.  $f(x) = 2 \log_2(x-5) + 4$  4.  $f(x) = \ln 3^{x-2}$   
 $f^{-1}(x) = \frac{2}{6}(x+6)$   $f^{-1}(x) = \frac{1}{2}(0.5^{x-4})$   $f^{-1}(x) = \frac{x+2}{\ln 3}$

5. A hurricane center uses the function  $s = 95 \log_d t + 75$  to relate the wind speed in miles per hour  $s$  and distance in miles  $d$  of a hurricane travels. How many miles will the hurricane travel with a wind speed of approximately 320 mi/h?  
 $320 = 95 \log_d t + 75$   $d \approx 10^{3.38} \approx 380$  mi

6. Which company's profit shows a greater average rate of change between 2010 and 2015?  
 Company A: \$1.5 million profit in 2010; after 5 years, grew exponentially to \$2.5 million.  
 Company B: profit, in million of dollars, modeled by  $P(t) = 1.3(1.15)^t$ , where  $x$  is the number of years after the end of 2010.  
 Company A: 0.2 million per year; Company B: about 0.26 million per year; Company B has greater average rate of change.

### Enrichment **O A**

Presents engaging problems and activities that extend the lesson concepts.

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### 6-4 Enrichment

Logarithmic Functions

According to the Richter scale, a magnitude +1.0 earthquake causes 10 times the ground motion of a standard earthquake. The Richter scale equation is  $R = \log\left(\frac{A}{A_0}\right)$ .  $A$  is the amplitude of a seismograph reading.  $A_0$  is the amplitude of a "standard earthquake", which is 1 micron, or  $10^{-6}$  cm.  $R$  is the magnitude of the earthquake on the Richter scale.

Answer the following questions.

- A scientist predicts there will be an earthquake with an epicenter in Palmdale, California and the magnitude of this earthquake will be an 8.2 on the Richter scale. For a seismograph 100 km from the epicenter, how many times as large will the amplitude of the reading be as  $A_0$ ?  
 $8.2 = \log\left(\frac{A}{A_0}\right)$   $\left(\frac{A}{A_0}\right) = 10^{8.2}$   $10^{8.2}$  times as large
- Earthquakes with magnitude greater than 5.0 may cause damage. A scientist calculates that if a seismograph is 100 km from an earthquake centered in San Diego, the amplitude of the reading will be  $10^{3.1}$  cm. Assuming she is right, what will the magnitude of the earthquake in San Diego be?  
 $R = \log\left(\frac{A}{A_0}\right) = \log\left(\frac{10^{3.1}}{10^{-6}}\right) = \log(10^{9.1}) = 9.1$
- Complete the following table to find out which city has the largest magnitude of the surface wave. The surface wave magnitude  $M$  of the earthquake is given by this formula:  $M = \log\left(\frac{S}{S_0}\right) + 1.66 \log D + 3.2$ .

| City          | (A, T, D)             | The surface wave magnitude of the earthquake |
|---------------|-----------------------|----------------------------------------------|
| San Francisco | $(10^{-1.1}, 5, 150)$ | 10.21                                        |
| San Diego     | $(10^{-1.1}, 5, 100)$ | 9.02                                         |
| Palmdale      | $(10^{-1.2}, 6, 180)$ | 10.47                                        |

a. What is the magnitude of the earthquake in San Francisco?  
 $R = \log\left(\frac{A}{A_0}\right) = \log\left(\frac{10^{-1.1}}{10^{-6}}\right) = \log(10^{4.9}) = 4.9$

b. Compare the magnitude of an earthquake that hits San Diego and an earthquake that hits San Francisco. Which city is hit with the most damage?  
 San Francisco

### Mathematical Literacy and Vocabulary **I O**

Helps students develop and reinforce understanding of key terms and concepts.

Name \_\_\_\_\_

### 6-4 Mathematical Literacy and Vocabulary

Logarithmic Functions

A jet takes off from an airport at sea level. Its altitude  $h$ , in feet, at time  $t$ , in seconds, is modeled by the function  $h = 1,650 \ln(t+1) + 100$ . How long will it take the jet to reach 10,000 ft?

Use the two sets of notes cards to find the solution of this problem. The set on the left explains the thinking. The set on the right shows the steps. Write the thinking and the steps in the correct order.

**Think Cards**

Divide each side by 1,650.

Write the original equation.

Subtract 100 from each side.

Write in exponential form.

Subtract 100 from each side.

Write the equation when the jet reaches 10,000 feet.

**Step Cards**

$h = 1,650 \ln(t+1) + 100$

$6 = \ln(t+1)$

$t \approx 402$  sec

$10,000 = 1,650 \ln(t+1) + 100$

$6 = \ln(t+1)$

$9,900 = 1,650 \ln(t+1)$

| Think                               | Write                                  |
|-------------------------------------|----------------------------------------|
| First, write the original equation. | Step 1 $h = 1,650 \ln(t+1) + 100$      |
| Second, substitute 10,000 for $h$ . | Step 2 $10,000 = 1,650 \ln(t+1) + 100$ |
| Third, subtract 100 from each side. | Step 3 $9,900 = 1,650 \ln(t+1)$        |
| Then, divide each side by 1,650.    | Step 4 $6 = \ln(t+1)$                  |
| Next, write in exponential form.    | Step 5 $e^6 = t+1$                     |
| Finally, subtract 1 from each side. | Step 6 $t \approx 402$ sec             |

### Digital Differentiated Resources

The Reteach to Build Understanding, Additional Practice, and Enrichment activities are available as digital assignments. These activities are automatically assigned when students complete the lesson quiz online and are automatically scored.

